Zhaoyu FAN

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Education

Zhejiang University · College of Computer Science and Technology · Hangzhou, China Sept. 2023 –Mar. 2026 (Expected)
Master of Science (M.S.) in Computer Science and Technology

- **Research Interests:** Multimodal Large Language Models, Natural Language Understanding, Knowledge Editing
- **Overall Average:** 93.44 / 100.0

Zhejiang University · College of Computer Science and Technology · Hangzhou, China Sept. 2019 - Jun. 2023
Bachelor of Engineering (B.E.) in Software Engineering

- **Overall GPA:** 3.92/4.0 (89.15/100); **GPA for the last two years:** 3.95/4.0 (89.47/100). **GPA Ranking:** Top 15%;

Publications

[1] **Fan, Z.**, Pan, K., Zhou, M., Qin, B., Li, J., Zhang, S., Zhang, W., Tang, S., Wu, F., & Zhuang, Y. (2025). Towards meta-cognitive knowledge editing for multimodal LLMs. *arXiv preprint arXiv:2509.05714*. Submitted to *The Web Conference (WWW) 2026* **(CCF-A, CORE A*)**.

[2] Pan, K.*, **Fan, Z.***, Li, J., Yu, Q., Fei, H., Tang, S., Hong, R., Zhang, H., & Sun, Q. (2024). Towards unified multimodal editing with enhanced knowledge collaboration. *Advances in Neural Information Processing Systems (NeurIPS)* **(CCF-A, CORE A*)**. **(Spotlight, Top 3.2% of submissions)**.
*Equal contribution

[3] Pan, K., Tang, S., Li, J., **Fan, Z.**, Chow, W., Yan, S., Chua, T.-S., Zhuang, Y., & Zhang, H. (2024). Auto-encoding morph-tokens for multimodal LLM. *International Conference on Machine Learning (ICML)* **(CCF-A, CORE A*)**. **(Spotlight, Top 4.6% of submissions)**.

[4] Huang, X., Zhang, P., Qin, B., **Fan, Z.**, Huang, H., Zhao, Y., Shi, S., Li, J., Zhuang, Y., & Tang, S. (2026). OmniMME: Decoupled multimodal capability evaluation for omni-language models. Under review at *AAAI* **(CCF-A, CORE A*)**.

[5] Qin, B., Zhang, P., Huang, X., **Fan, Z.**, Huang, H., Zhao, Y., Li, J., Tang, S., & Zhuang, Y. (n.d.). Modality expansion for multimodal large language models via distribution and magnitude aware model merging. Under review at *IEEE Transactions on Multimedia* **(CCF-A, JCR Q1)**.

Research Experience

Multimodal Knowledge Editing for Large Language Models · *Zhejiang University* Jan. 2024 - Present

- Co-developed UniKE for multimodal knowledge editing, solving the problem that existing intrinsic knowledge editing and extrinsic knowledge retrieval methods have difficulty balancing multiple metrics when applied to MLLMs. Designed a novel architecture using additional neurons to store intrinsic knowledge and pre-trained parameters to store extrinsic knowledge. (Research accepted at NeurIPS 2024 as Spotlight paper.)
- Identified limitations in existing cognitive-based editing methods and proposed the CogEdit benchmark to comprehensively measure the metacognitive knowledge editing ability of MLLMs. Developed the MIND framework to achieve more accurate and stable metacognitive knowledge editing, enabling the model to determine when to replace outdated knowledge, when to apply new information, and how to handle noise during the editing process. (Research available at arXiv.2509.05714, submitted to The Web Conference (WWW) 2026.)

Hallucination Mitigation for Multimodal Models · *Zhejiang University* Mar. 2025 - Present

- Designed a benchmark to evaluate hallucinations in Multimodal Large Language Models by introducing structured knowledge conflicts, enabling robust assessment of model reliability under contradictory information.
- Proposed a hallucination mitigation framework based on conflict detection and cross-model information transfer, achieving preliminary results; extended validation and manuscript preparation are in progress (Planned submission to ACL 2026).

Alibaba Group · *Alibaba Holding - AGI Safety Lab - Research Intern* Jun. 2025 - Present

- Designed a novel self-play(adversarial) based continue pre-training method to inject domain-specific knowledge into foundation models while preserving general capabilities, achieving substantial gains on 20+ risk-domain benchmarks of the Qwen2-VL family, especially in security auditing tasks. This method will be incorporated into the training of the next-generation foundation model.

- The approach is being integrated into the next-generation foundation model; a related patent application titled "**A Model Knowledge Injection Method Based on Self-Play**" has been submitted and a manuscript is in preparation for ICML 2026.

Cross-Modal Model Fusion Research · *Huawei Cloud* Jan. 2025 - Sept. 2025

- Contributed to the construction of an ability-centric multimodal evaluation benchmark (image, video, audio, 3D), enabling fine-grained assessment of model capabilities across modalities; conducted validation experiments on multiple state-of-the-art models (Submitted to AAAI 2026).
- Investigated novel cross-modal model fusion strategies for image and video tasks, demonstrating superior effectiveness over conventional continual learning approaches (Submitted to IEEE Transactions on Multimedia).

Persona-based Knowledge Transfer for Large Language Models · *Undergraduate Thesis* Sept. 2022 - Apr. 2023

- Proposed a novel persona-based instruction tuning paradigm where an LLM adopts designed personas (e.g., "Logician," "Linguist") to generate specialized data, enabling controlled knowledge transfer to smaller models.
- Achieved a 3.44% overall improvement and up to 423% enhancement in causal reasoning tasks on the Super-Natural Instruction dataset with the LLaMA-7B model.

Zhejiang University Supercomputing Team(ZJUSCT) · *Zhejiang University* Sept. 2020 - Jun. 2023

- Main focus is on Python performance optimization, solving the efficient operation of LLM/scientific computing programs when computing resources are limited.
- Participated in the world ASC Student Supercomputer Challenge ASC20-21, ASC22, and was responsible for the LLM distributed computing competition in the competition. Followed the school to Shenzhen to participate in the finals in ASC20-21, and as the captain of the second team of Zhejiang University in ASC22, led the team to win the second prize.

Internships

Mjoys Technology · *Zhihai Jinpan - Financial Vertical Domain Large Language Model* Aug. 2023 - May 2024

- To address the issues of professionalism and data security compliance of general large models in the financial field, a privately deployable financial vertical domain model was built. Its knowledge covers financial knowledge graphs, financial texts, financial dialogues, etc. The model has been put into production as an intelligent assistant.
- Responsible for the construction of distributed computing related code, corpus organization, instruction fine-tuning, model deployment, and writing related documents.

NetEase, Inc. · *NetEase Fuxi Lab - Fuxi Robot* Oct. 2022 - Feb. 2023

- Spearheaded the development, maintenance, and performance tuning of an in-house, Python-based communication SDK for model training, ensuring the efficiency and stability of training tasks.
- Contributed to performance optimization and feature iterations, ensuring the SDK delivered robust serialization/deserialization functionalities while maintaining performance comparable to mainstream frameworks like Google Protobuf.

Awards and Honors

Award of Honor for Graduate, Zhejiang University **(University-level)**	2023-2024; 2024-2025
Graduate with Merit A Performance, Zhejiang University **(University-level)**	2023-2024; 2024-2025
The Second Class Prize, ASC22 Student Supercomputer Challenge **(Captain, International-level)**	2022
Excellent Engineer Scholarship **(University-level)**	2021-2022
Yongping Scholarship **(University-level)**	2021-2022
Zhejiang University Scholarship (Third Prize) **(University-level)**	2021-2022
Honorable Mention, MCM/ICM 2021 **(International-level)**	2021
Zhejiang University Scholarship (Second Prize) **(University-level)**	2019-2020; 2020-2021
First Prize, National College Student Mathematics Contest **(National-level)**	2021
First Prize, Physics Innovation Competition **(National-level)**	2021

Skills

- **Programming Languages**: Python, C/C++, Java, Shell, Java, SQL.
- **Techniques**: Pytorch; Distributed Computing (MPI, NCCL); Linux kernel developer;